

Proteoform Profiles Distinguish Adult Zebrafish Telencephalon and Optic Tectum

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Abstract

The adult zebrafish telencephalon and optic tectum serve different neural functions: one is associated with forebrain integration, memory, and social behavior, the other with visual orientation and sensorimotor control. Top-down proteomics now makes that division visible at the level of intact proteoforms. Using the region-resolved LCM-CZE-MS/MS dataset of [Lubeckyj and Sun \(2022\)](#) and the accompanying PRIDE Tel2 and Teo2 spreadsheets, we found a strikingly sparse shared proteoform set: only 35 of 807 article-reported proteoforms were shared, and strict accession+proteoform matching gave 29 shared exact IDs among 805. The separation had biological direction. Telencephalon favored neuropeptide and synaptic entries, whereas optic tectum favored reticulon-, actin-, myosin-, and troponin-linked entries; an independent marker panel recovered the same regional axis. Sensitivity analyses for uneven recovery, unequal identification counts, identifier error, and protein-level collapse preserved the low-overlap result. By contrast, the apparent excess of N-terminal acetylation in telencephalon weakened once differential recovery was modeled. These data show that two anatomically and functionally distinct zebrafish brain regions also occupy sharply distinct proteoform states.

1 Introduction

Adult zebrafish telencephalon and optic tectum solve different biological problems. The telencephalon participates in social behavior, learning, memory, and other forebrain functions ([Stednitz et al., 2018](#); [Reemst et al., 2023](#); [Pandey et al., 2023](#)). The optic tectum converts visual input into orienting and avoidance behavior ([Roeser and Baier, 2003](#); [Orger, 2016](#)). If this division reaches down into molecular organization, it should be especially visible at the proteoform level, where sequence variants and post-translational modifications remain distinct instead of being merged into protein totals.

The LCM-CZE-MS/MS study of [Lubeckyj and Sun \(2022\)](#) makes the question unusually concrete. It reported hundreds of proteoforms from adult zebrafish telencephalon and optic tectum and connected those profiles to regional function. Its PRIDE Tel2 and Teo2 spreadsheets add something further: they expose which proteoforms are actually shared, which are region-restricted, and how those patterns change when entries are matched as exact accession+proteoform identities or collapsed to broader protein units.

Read directly, the proteoform layer gives a clear answer. Telencephalon and optic tectum share very few proteoforms. The entries that distinguish them fall along a recognizable biological axis: neuropeptide and synaptic proteins on the telencephalic side, reticulon- and cytoskeleton-linked proteins on the tectal side. The modification layer is more guarded; an apparent telencephalic excess of N-terminal acetylation weakens once uneven recovery is taken into account. The main discovery is therefore not a global PTM program, but a strong regional separation in intact proteoform state.

The result matters beyond one dataset. Prior functional studies and atlases establish the broad biology of these regions ([Stednitz et al., 2018](#); [Pandey et al., 2023](#); [Roeser and Baier, 2003](#); [Orger, 2016](#)), and recent top-down work shows that proteoform-level structure in zebrafish brain remains informative beyond a single pilot study ([Askarani et al., 2026](#)). What has been missing is an explicit view of regional overlap

and separation at intact-proteoform resolution. The evidence here supports a simple conclusion: the adult zebrafish telencephalon and optic tectum are not merely different anatomical samples of the same proteome. They are distinguished by different proteoform states that track their different roles in the brain.

2 Materials and Analysis

2.1 Source material

We studied the adult zebrafish telencephalon and optic-tectum LCM-CZE-MS/MS dataset of [Lubeckyj and Sun \(2022\)](#). The analysis used the article, its supplementary workbook, and the two spreadsheets deposited in the PRIDE proteomics repository, `Proteoform_ID_Tel2.xlsx` and `Proteoform_ID_Teo2.xlsx`. The supplement reports TopPIC identification at 1% proteoform-spectrum-match FDR and 5% proteoform-level FDR, with duplicate label-free quantification in TopDiff. The `Tel2` and `Teo2` sheets embedded in the supplement match the PRIDE files exactly, so the region spreadsheets served as the quantitative source and the supplement supplied acquisition and search context. Because processed feature histories are not included, we analyzed the processed proteoform spreadsheets directly.

The primary comparison unit was the accession-plus-proteoform pair. This is stricter than comparing proteoform strings alone, because the same peptide-like string can occur under more than one accession. Accession identity was therefore retained in the main overlap analysis, while protein and gene-symbol collapse were used as sensitivity views. Marker families were locked to the biological language of the original study: exact gene-symbol matches for NPY, PENKB, PYY-A, and neurogranin a, and description-keyword matches for synucleins, reticulon, myosin, actin, and troponin. The mapping rules, membership table, and traceability record are generated locally by `results/analyze_region_proteome.py`.

2.2 Derived metrics

Three questions guided the analysis. First, how much of the proteoform repertoire is shared between the two regions? Second, do region-enriched proteoforms follow known telencephalic and tectal biology? Third, does the apparent N-terminal acetylation difference survive adjustment for unequal recovery? Regional separation was assessed with article-level counts, exact accession+proteoform overlap, canonicalized discrepancy diagnostics, protein-level collapse, abundance-weighted similarities, fixed-margin overlap tests, count-normalized subsampling, a duplicate-based detectability correction, and conservative misidentification bounds keyed to the published FDR values. Biological direction was assessed with the marker families emphasized in the original report and with an independent literature-derived composition panel. The acetylation analysis was restricted to matched marker subsets and then stress-tested against simple proxies for unequal recovery.

Robustness checks followed the structure of the available files: gene-symbol collapse, alternative abundance normalizations, duplicate-informed missingness bounds, a minimal MNAR-style low-intensity fill, family-size-preserving randomization, conservative spillover reassignment, motor-family exclusion, protein collapse, and rarefaction of the larger optic-tectum list down to the telencephalon count. The duplicate-based overlap correction was repeated after stratifying exact IDs by precursor-mass tertiles and by mean-intensity tertiles within each region. Acetylation was examined with a coarse logistic sensitivity model in both orientations: region as the outcome with acetylation as a predictor, and acetylation as the outcome with region as a predictor. Both models used log precursor mass, sequence span length, and first-residue position as covariates, followed by a restricted analysis on rows whose first residue lay at or within two amino acids of the N terminus. Formulas and implementation details are given in the Appendix.

2.3 Interpretive scope

The available material supports overlap analysis, marker-family mapping, and conservative PTM sensitivity checks. It does not support a full feature-level error model. The strongest claims are therefore about regional separation and the biological direction of that separation. On those questions, the evidence is direct.

3 Results

3.1 Telencephalon and optic tectum are sharply separated at the proteoform level

At the proteoform level, the two regions separate cleanly. The article reported 309 telencephalic proteoforms and 533 tectal proteoforms, with only 35 shared between them. Exact accession+proteoform matching of the PRIDE Tel2 and Teo2 spreadsheets is even stricter, yielding 29 shared entries among 805 total exact IDs. Protein-level overlap is broader, but still modest. Even after collapsing to gene symbols, the overlap rises only to a Jaccard value of 0.2302, far below the fixed-margin independence baseline (centered Jaccard -0.2034 , $p = 6.29 \times 10^{-28}$). Telencephalon and optic tectum are therefore not minor variants of the same local proteoform pool.

The separation persists across encoding choices. Abundance weighting changes the overlap only slightly: across five normalization schemes, weighted Jaccard remains between 0.0427 and 0.0557, while Bray–Curtis similarity remains between 0.0820 and 0.1055. Fixed-margin testing places the observed exact-ID overlap far below the overlap expected from the regional list sizes alone (expected Jaccard 0.3152, 95% null interval 0.2890–0.3430, centered Jaccard -0.2792 , $p = 5.56 \times 10^{-183}$). The difference between the article’s 35 shared proteoforms and the spreadsheet-derived exact-ID count of 29 appears to be representational rather than biological: relaxed matching that strips bracketed PTM annotations and punctuation while keeping accessions fixed raises the shared count to 44, but the overlap remains very low (canonicalized Jaccard 0.0615, centered Jaccard -0.2627 , $p = 5.01 \times 10^{-143}$). The biological picture is the same in every case.

Sensitivity analysis reaches the same end point. Duplicate-informed missingness bounds, jackknife estimates, and minimal MNAR-style fills never turn the dataset into a high-overlap case. In a simple occupancy model built from duplicate rediscovery, correcting for uneven recovery raises the exact-ID Jaccard only from 0.0360 to 0.0432. Stratifying the same correction by precursor-mass tertiles or mean-intensity tertiles changes the adjusted Jaccard only to 0.0451 and 0.0445. Subsampling the larger optic-tectum list down to the telencephalon count gives a median exact-ID Jaccard of 0.0289, with a 95% sampling interval of 0.0185–0.0377; canonicalized and protein-collapsed views remain low as well. Conservative identifier-error bounds are equally modest: even if 5% of region-unique exact IDs are treated as possible false discoveries in the direction most favorable to overlap, the Jaccard ceiling rises only to 0.0540. Uneven detectability, unequal list size, and plausible identifier error soften the edge of the split, but they do not erase it.

3.2 Marker families follow known regional biology

Low overlap by itself could still be biologically uninformative. Here it is not. The curated marker panel is concentrated almost entirely in the region expected from known biology. Telencephalon-associated markers contribute 23 matched counts and only 2 spillover counts, whereas optic-tectum-associated markers contribute 102 matched counts and 1 spillover count. Across the full panel, 125 of 128 observations fall in the expected region. Because optic tectum contributes the larger raw proteoform total in the original study, this alignment is not a trivial consequence of abundance imbalance.

The statistical support is correspondingly strong. The panel yields an odds ratio of 1173 with an approximate 95% confidence interval from 102.0 to 13,495.1 and a one-sided exact p value of 5.12×10^{-22} . A family-size-preserving randomization reaches the same conclusion without relying on asymptotic logic: under

Table 1: Core regional separation metrics. The first two rows preserve the article-level summary, and the remaining rows use the PRIDE region spreadsheets plus detectability and identifier-error bounds.

Metric	Value
Article shared proteoforms	35
Article Jaccard overlap	0.0434
Exact-ID shared proteoforms	29
Exact-ID Jaccard overlap	0.0360
Weighted Jaccard overlap	0.0427
Protein overlap fraction	0.2151
Count-normalized exact-ID Jaccard median	0.0289
Occupancy-adjusted shared exact IDs	43.2
Occupancy-adjusted Jaccard overlap	0.0432
5% misidentification ceiling	0.0540

Table 2: Curated marker families used in the axis-alignment analysis. Exact accession+proteoform members for each family are listed in the local marker-membership file, but the biological interpretation still follows the marker groups named in the original paper.

Marker group	Tel	Teo	Interpretation
Pro-neuropeptide Y	1	0	feeding-related neuropeptide signaling
Synenkephalin	1	0	affective signaling
Peptide YY-A	1	0	peptide hormone signaling
Neurogranin a	6	0	memory-linked synaptic signaling
Synucleins total	14	2	synaptic and mnemonic family
Reticulon	0	4	retinotectal organization
Myosin	0	48	motor machinery
Actin	0	19	cytoskeletal movement
Troponin	1	31	contraction-linked regulation

200,000 randomizations that preserved the nine family sizes and the global 25-versus-103 telencephalon/optic-tectum label totals, none matched the observed 125 correctly placed counts. All nine curated marker groups point in the expected direction, and the signal remains strong after protein collapse and intensity weighting. The separating proteoforms trace a coherent biological axis.

The motor-heavy families require a narrow interpretation. A detailed accession breakdown shows that 86.3% of the myosin and troponin observations carry skeletal- or cardiac-muscle-like annotations. They therefore cannot anchor the regional biology on their own. Their removal is more informative: after excluding myosin, actin, and troponin altogether, the matched-region fraction remains 0.9310.

The same pattern survives more stringent perturbations. Leave-one-out removal of any single marker family keeps the matched-region fraction between 0.9625 and 0.9911. Conservative spillover reassignment still leaves the panel strongly aligned, and removing the motor-associated myosin, actin, and troponin families leaves an alignment fraction of 0.9310. A simple purity screen points in the same direction: the region spreadsheets contain only two optic-tectum glial-support sentinels but 23 telencephalon myelin sentinels versus three in optic tectum.

An independent panel curated from the broader zebrafish literature reaches the same conclusion. Adult zebrafish myelin markers and optic-tectum radial-glia markers yield 32 matched counts and only 3 spillover

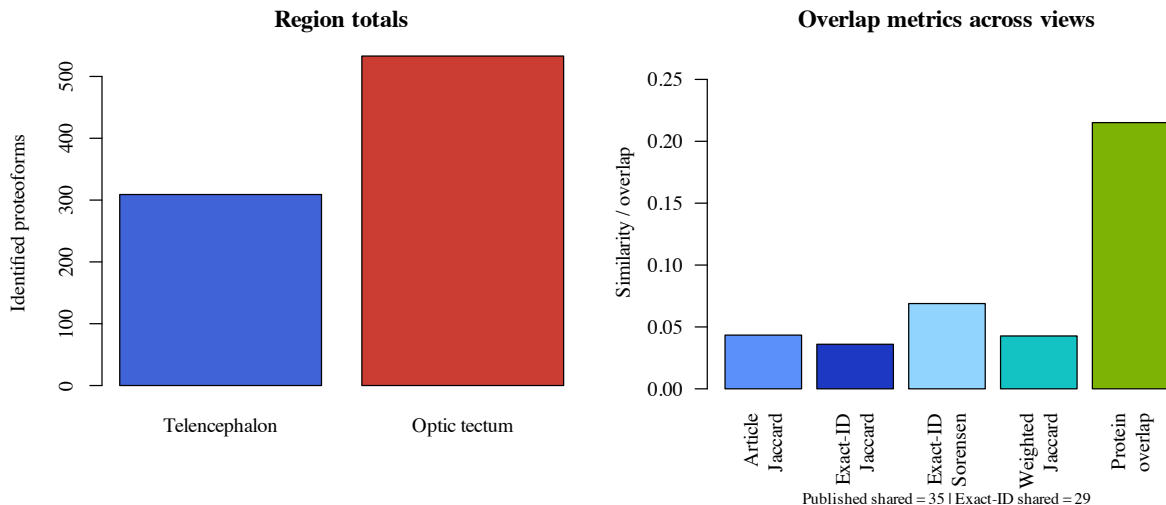


Figure 1: Regional totals and overlap metrics. The PRIDE region spreadsheets contain 303 exact IDs in telencephalon and 531 in optic tectum, with 29 exact-ID overlaps. This spreadsheet-based count is slightly stricter than the article aggregate, but both support the same low-overlap picture.

Table 3: Axis-level alignment compared with the naive baseline implied by regional prevalence.

Functional axis	Observed	Wilson 95% CI	Baseline	Lift
Telencephalon-associated	0.9200	0.7503–0.9778	0.1875	0.7325
Optic-tectum-associated	0.9903	0.9470–0.9983	0.8125	0.1778

counts, for an alignment fraction of 0.9143 and a one-sided exact p value of 6.68×10^{-4} . The panel is smaller and composition-oriented, but it widens the result beyond the original functional vocabulary. Count-normalized subsampling also leaves the original marker axis intact: when the larger optic-tectum row set is sampled down to the telencephalon row count, the median marker-alignment fraction is 0.9756.

3.3 The PTM layer is interesting, but the acetylation signal is not yet stable

The PTM layer adds a useful boundary to the discovery. The original article reported 807 proteoforms in total, 358 with mass shifts and 211 with N-terminal acetylation, so the dataset is plainly proteoform-rich rather than a protein list in another format. Yet the specific acetylation contrast is less secure than the regional overlap and marker results. In the matched marker subsets, 14 of 23 telencephalon-associated proteoforms are N-terminally acetylated, compared with 16 of 102 optic-tectum-associated proteoforms. In a coarse logistic model with region as the outcome and acetylation, log precursor mass, sequence span length, and first-residue position as predictors, the effect weakens markedly. The adjusted odds ratio falls to 2.94 (95% CI 0.70–12.30; $p = 0.140$), and the first-residue ≤ 2 subset shows no enrichment. Reversing the model makes the same point: with acetylation as the outcome and region as the main predictor, the regional odds ratio is 0.26 with $p = 0.180$. Acetylation remains a lead, not a settled regional program.

The duplicate analyses help explain why a small study can reveal a clear regional structure. Duplicate runs recovered a mean of 232 proteoforms in telencephalon and 356 in optic tectum, corresponding to recovery fractions of 0.7508 and 0.6680 relative to the regional totals. The design is limited, but the regional split is not a single-run accident.

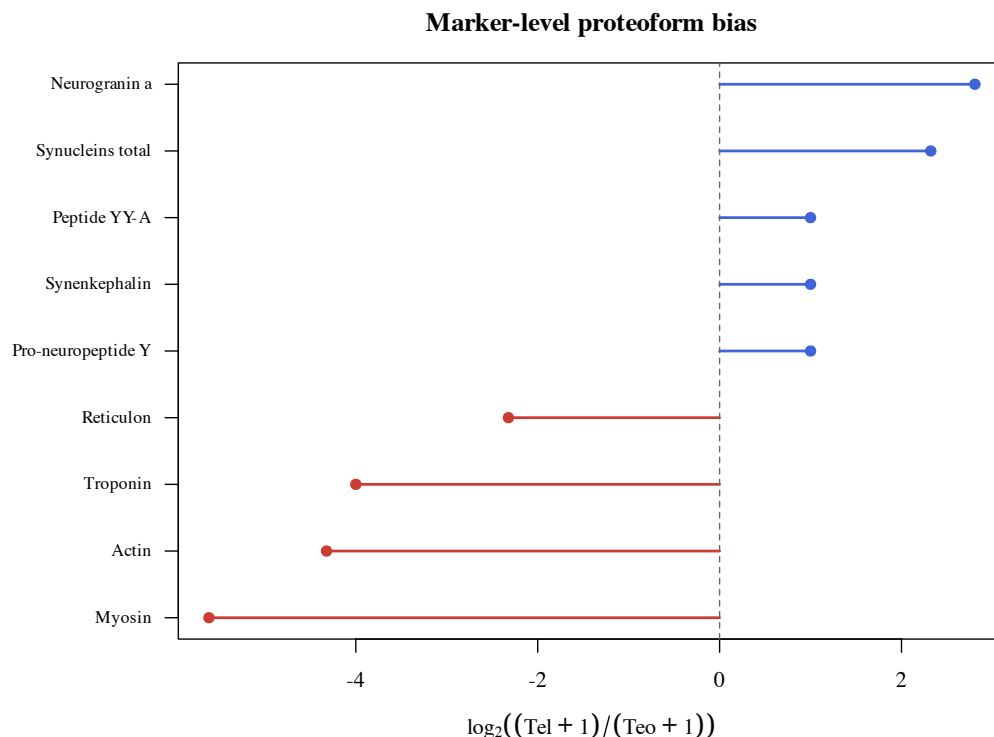


Figure 2: Marker-level proteoform bias across 128 curated proteoform observations in nine marker families. Positive values favor telencephalon and negative values favor optic tectum.

4 Discussion

The adult zebrafish telencephalon and optic tectum carry different proteoform complements. The evidence is convergent: sparse exact-ID overlap, coherent marker-family direction, and conservative sensitivity analyses that never turn the two regions into a high-overlap pair.

The biology of the split is as important as its magnitude. Telencephalon is enriched for neuropeptide and synaptic entries consistent with a forebrain region devoted to integration, learning, and behavioral regulation. Optic tectum is enriched for reticulon-, actin-, myosin-, and troponin-linked proteoforms consistent with retinotectal circuitry and the cellular architecture needed for visually guided orienting. The two regions differ in a way that remains intelligible against established regional function.

The motor-associated families deserve a narrow reading. Most of the myosin and troponin accessions in the region spreadsheets are skeletal-muscle-like annotations, which means they cannot carry the biological interpretation by themselves. Their removal is the useful test. Once those families are excluded, the remaining panel still points strongly in the same direction, and the independent literature-derived panel does as well.

The larger tectal identification count also cannot explain the result. When the optic-tectum list is repeatedly sampled down to the telencephalon count, exact-ID overlap remains sparse and the marker axis remains almost unchanged. The simple composition guardrails point the same way: myelin sentinels favor telencephalon, optic-tectum glial-support sentinels are few, and the motor-heavy families are dispensable for the main alignment. Direct purity measurements would still be valuable, but the available checks do not support a contamination-driven explanation.

That regional logic would be easy to blur in a protein-collapsed analysis. Protein overlap is substantially broader than proteoform overlap, which means that much of the distinction lies in the resolved proteoform

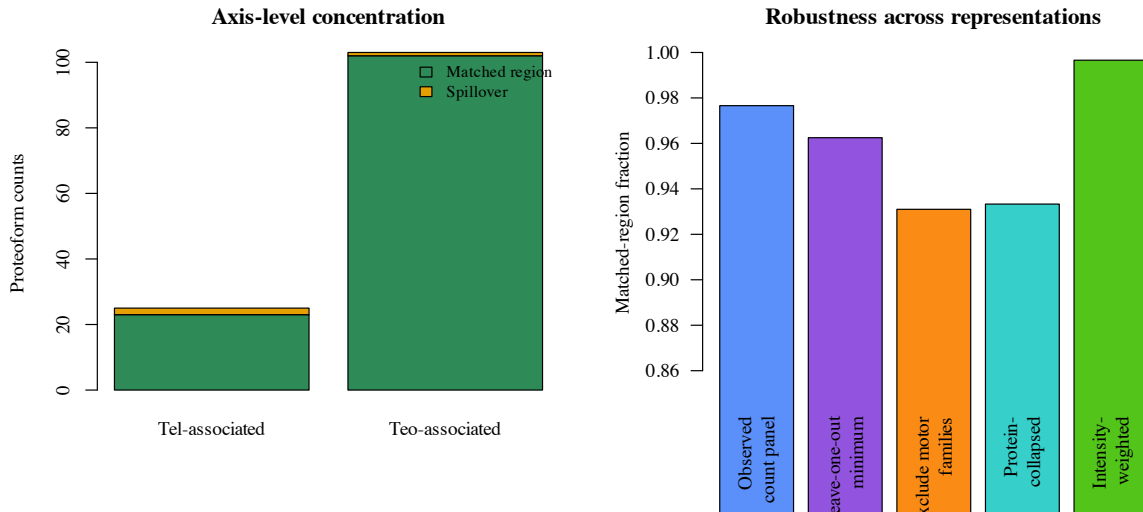


Figure 3: Axis-level concentration and robustness summaries for the nine curated marker families. The overall alignment signal remains high after leave-one-out perturbation, after excluding motor families, after collapsing to proteins, and after weighting by mean duplicate intensity.

Table 4: Sensitivity checks requested by the external review. The motor-family exclusion uses a Haldane correction because one spillover cell is zero after exclusion.

Scenario	Alignment	Odds ratio	One-sided p
Observed panel	0.9766	1173.0	5.12×10^{-22}
Shift 1 matched count per axis	0.9609	370.3	2.01×10^{-19}
Shift 2 matched counts per axis	0.9453	175.0	3.73×10^{-17}
Shift 3 matched counts per axis	0.9297	99.0	3.93×10^{-15}
Exclude myosin, actin, troponin	0.9310	84.6	6.32×10^{-4}

state rather than in simple protein presence or absence. For a region-resolved nervous system dataset, that is the central biological lesson. The proteoform layer preserves a regional texture that broader summaries dilute.

The independent composition panel supports the same conclusion from a different angle. Myelin markers remain associated with telencephalon, and optic-tectum radial-glia markers remain associated with optic tectum. The panel is small, but it matters because it shows that the result does not disappear when the vocabulary is widened beyond the original paper’s favored examples.

The result also clarifies the place of microdissected top-down proteomics. Imaging-based approaches such as LAESI and nano-DESI MSI bring direct spatial fidelity and on-tissue proteoform mapping (Kiss et al., 2014; Yang et al., 2022). LCM-CZE-MS/MS makes a different bargain: it sacrifices within-section imaging for deeper coverage within anatomically defined regions. That depth is what makes the overlap structure and marker architecture legible here.

The region spreadsheets also reveal a practical lesson about proteoform analysis. The gap between the article’s 35 shared proteoforms and the strict exact-ID count of 29 appears to arise from representation rather than biology. Relaxed matching changes the count, but not the inference. Standards such as ProForma matter because they help separate notational variation from biological difference (LeDuc et al., 2018, 2022). More broadly, current proteomics discussions have stressed that data-analysis choices, missingness handling, and provenance shape the conclusions readers can trust (Vanderaa and Gatto, 2023). These data do not contain

Table 5: Technical sensitivity values reported in the original study and extended with exact-ID duplicate-incidence counts from the PRIDE region spreadsheets.

Region	Mean	SD	CV	Recovery	exact-ID q_1	Chao2
Telencephalon	232	28	0.1207	0.7508	152	385.5
Optic tectum	356	88	0.2472	0.6680	351	887.0

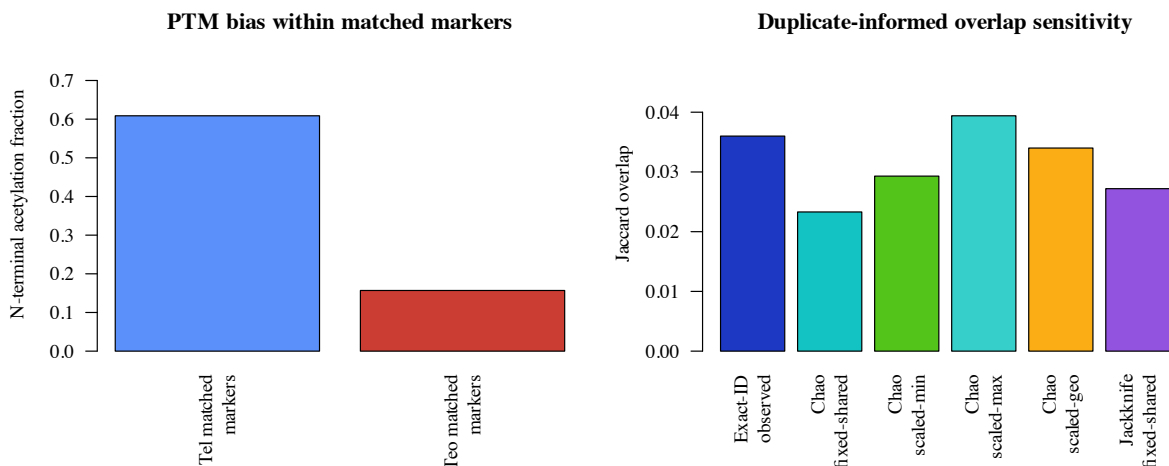


Figure 4: PTM and missingness-sensitive follow-up analyses. Left: N-terminal acetylation in the matched-marker subsets (23 telencephalon counts and 102 optic-tectum counts). Right: duplicate-informed Chao2 sensitivity changes the exact-ID Jaccard modestly, but never enough to erase the regional separation.

enough localization metadata for full ProForma 2.0 serialization of every entry, but they contain enough structure to show that the regional conclusion is stable across reasonable canonicalization choices.

The same evidence illustrates a deeper interpretive point about proteoform families. Exact accession-resolved entries, protein collapse, and gene-symbol collapse do not tell the same biological story. Coarser identifiers broaden overlap, yet the regions remain clearly separated even after that coarsening. This is why proteoform-family thinking matters in top-down proteomics: biological meaning is lost step by step as the identifier is relaxed (Schaffer et al., 2018).

The detectability and identifier-error analyses sharpen the same point. Correcting for uneven recovery raises the exact-ID Jaccard only modestly, and the result hardly changes when the same correction is stratified by mass or by intensity. Conservative error bounds do not drive the two regions toward a high-overlap case either. To overturn the result, one would need far more hidden sharing than the duplicate structure or the published FDR values make plausible.

By contrast, the acetylation result softens as soon as detectability is modeled explicitly. That boundary is useful. It marks the point at which the data stop warranting a strong biological claim. The durable finding is regional proteoform separation, not a settled regional PTM program.

4.1 Limits of inference

The claim is shaped by the available material: processed public tables rather than raw feature histories. The evidence is strongest where the files are most direct: regional separation and the biological direction of that

separation. The duplicate-labeled spreadsheets support missingness bounds, normalization checks, marker traceability, occupancy-style detectability correction, and a limited PTM screen, but they do not expose the feature histories, localization confidence, or cross-region replicate structure that a raw TopPIC export would provide (Choi and Liu, 2022). Those limits narrow the claim. They do not weaken the central result.

The strongest extension would combine public raw files with processed search intermediates so that the same question could be asked at feature level and under richer missingness models (Karpievitch et al., 2012; Lazar et al., 2016). Additional brain regions would test how general this axis of separation is, and explicit purity measurements would strengthen the anatomical interpretation. This dataset already secures the main point: adult zebrafish telencephalon and optic tectum are distinguished by different proteoform profiles, and those profiles track known regional biology.

5 Conclusion

Adult zebrafish telencephalon and optic tectum are separated by distinct proteoform profiles, not merely by small shifts in abundance. The separating proteoforms follow the expected biology of forebrain integration on one side and visuomotor architecture on the other, whereas the apparent acetylation asymmetry remains provisional. The proteoform layer therefore preserves a biological order that protein-level summaries partly obscure. In these data, two neighboring parts of the adult brain carry different molecular signatures of what they do.

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A Technical Notes

A.1 Overlap and representation

The primary overlap unit is accession plus proteoform string. This preserves cases in which similar proteoform-like strings are assigned to different accessions. Protein-level and gene-symbol summaries are reported as coarser views, not as replacements for the accession-resolved analysis.

Canonicalized matching was used only to diagnose representation effects. In that view, bracketed PTM annotations, punctuation, and non-sequence characters were removed while accession identity was retained. The canonicalized comparison explains why the article-level count of 35 shared proteoforms lies between the strict exact-ID count and the relaxed representation, but it does not change the regional inference.

Prevalence-adjusted overlap was evaluated with fixed telencephalon and optic-tectum margins. Let n_T and n_O be the two regional list sizes, N the union size, and X the observed shared count. The observed Jaccard is $J = X/(n_T + n_O - X)$. The marginal-prevalence baseline is

$$J_0 = \frac{p_T p_O}{p_T + p_O - p_T p_O},$$

where $p_T = n_T/N$ and $p_O = n_O/N$. The centered Jaccard is $J - J_0$. For uncertainty under fixed margins, the shared count was treated as $X \sim \text{Hypergeometric}(N, n_T, n_O)$, and the hypergeometric support was transformed to Jaccard values $x/(n_T + n_O - x)$. Under strict exact-ID matching, the observed Jaccard was 0.0360 against an independence baseline of 0.3152 and a 95% fixed-margin null interval of 0.2890–0.3430. Under canonicalized matching, the observed Jaccard was 0.0615 against a baseline of 0.3243. After protein collapse, the overlap rose to 0.2151 but still remained below the protein-level baseline of 0.4280.

A.2 Detectability and missingness

Duplicate `Match/No match` flags in the PRIDE region spreadsheets supplied the incidence counts used for Chao2 and first-order jackknife richness calculations. Blank intensity cells were treated as absent detections in the baseline analysis. A minimal MNAR-style sensitivity filled missing exact IDs to half of the region-specific 5th- or 10th-percentile nonzero intensity.

The duplicate-based detectability correction treated each exact ID as detected in one run or both runs within a region. The resulting two-run model estimates a region-specific per-run detection probability and an “observed at least once” probability. Shared exact IDs were then adjusted by the product of the two region-level detection probabilities. The same calculation was repeated after stratifying exact IDs by precursor-mass tertiles and by mean-intensity tertiles.

Misidentification sensitivity asked how much the exact-ID Jaccard could rise if an equal fraction of region-unique exact IDs in both regions were paired representation or identification errors. The 1% and 5% scenarios follow the PrSM- and proteoform-level FDR values reported in the supplement. Even the 5% scenario left the overlap low.

Identification-count rarefaction asked whether the larger optic-tectum list alone created the separation. In each of 10,000 trials, the larger regional set was sampled without replacement down to the smaller set size, with the seed fixed at 20260515. The calculation was repeated for exact accession+proteoform IDs, canonicalized accession+sequence IDs, protein accessions, and row-level marker alignment. The output is written to `results/identification_count_rarefaction.csv`.

A.3 Marker families and acetylation

Marker alignment was evaluated with matched-versus-spillover counts, Wilson intervals, an approximate odds-ratio interval, an exact directional test, and family-size-preserving randomization. The randomization preserved the nine family sizes and the global telencephalon/optic-tectum label totals while shuffling labels across panel counts. Leave-one-family-out analysis, motor-family exclusion, protein collapse, and intensity weighting all retained the same regional direction.

The acetylation analysis used matched marker rows only. One logistic model coded region as the outcome and used acetylation status, log precursor mass, sequence span length, and first-residue position as predictors. The inverse model coded acetylation as the outcome and region as the main predictor with the same covariates. A second analysis restricted rows to first residue ≤ 2 as a stricter N-terminal screen. These checks weakened the apparent telencephalic acetylation enrichment.

A.4 Limits of the public files

The journal supplement records that the region spreadsheets were generated with 1% PrSM FDR and 5% proteoform-level FDR in TopPIC and quantified across duplicate runs in TopDiff. The PRIDE landing page lists the four raw duplicate files, but the processed feature histories needed for a full raw-data rerun are not present. The analysis is therefore strongest for regional overlap, marker-family direction, and conservative PTM sensitivity; it is not a complete proteoform atlas of the adult zebrafish brain.

All derived tables in this manuscript are produced by `results/analyze_region_proteome.py`. The compact reproduction notebook is `code/reproduce_tables_and_figures.ipynb`. Canonicalization examples, marker-family memberships, overlap tests, and the source-to-claim traceability record are saved under `results/` and `data/evidence_traceability.json`.